

ECON 21030 HW1

ANDREW LYS

1. PROBLEM 1

Let

$$M_X(t) = E[\exp(tX)]$$

whenever this expectation exists in a neighborhood of $t = 0$.

Problem 1.a. Write down and simplify as much as possible the MGF for a normal distribution with mean μ and variance σ^2 .

Proof. Let $X = \mu + \sigma Z$, where $Z \sim N(0, 1)$. We know that

$$\exp\left(-\frac{1}{2}\sigma^2 t + \sigma B_t\right)$$

is a martingale, so we know that

$$\exp(\sqrt{t}\mu) = E \exp\left(-\frac{1}{2}t\sigma^2 + \sigma\sqrt{t}Z + \mu\sqrt{t}\right) = E \exp\left(-\frac{1}{2}t\sigma^2 + \sqrt{t}X\right)$$

Which implies that

$$M_X(t) = \exp\left(t\mu + \frac{1}{2}t^2\sigma^2\right)$$

□

Problem 1.b. What are the first and second derivatives of the MGF for this normal random variable at $t = 0$? How do they relate to its moments?

Proof. We have that

$$M_X^{(n)}(t) = E[X^n \exp(tX)]$$

so that

$$M_X^{(n)}(0) = E[X^n]$$

In particular, we have

$$M_X'(0) = \mu \quad M_X''(0) = \sigma^2 + \mu^2$$

□

Problem 1.c. Take a third order Taylor expansion of $\exp(tX)$ around zero. What is its expectation?

Proof.

$$\exp(tX) = 1 + tX + \frac{t^2 X^2}{2} + \frac{t^3 X^3}{6} + O(t^4 X^4)$$

Taking expectations, we get

$$E \exp(tX) = 1 + tE[X] + \frac{t^2}{2}E[X^2] + \frac{t^3}{3!}E[X^3]$$

This leads us to believe that

$$M_X^{(n)}(0) = E[X^n]$$

□

Problem 1.d. Consider i.i.d. random variables $X_1, X_2, \dots$ with finite mean μ and variance σ^2 . Define

$$Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{X_i - \mu}{\sigma}$$

How does the MGF of Z_n relate to the MGF of $Y_i = \frac{X_i - \mu}{\sigma}$?

Proof. We have that

$$\begin{aligned} M_{Z_n}(t) &= E \exp \left(\frac{t}{\sigma \sqrt{n}} \sum_{i=1}^n (X_i - \mu) \right) \\ &= \prod_{i=1}^n E \exp \left(\frac{t}{\sigma \sqrt{n}} (X_i - \mu) \right) \\ &= \prod_{i=1}^n \exp \left(\frac{1}{2} \left[\frac{t}{\sigma \sqrt{n}} \right]^2 \sigma^2 \right) \\ &= \exp \left(\frac{1}{2} t^2 \right) \end{aligned}$$

This is precisely the MGF of Y_i

□